

LENGTH-ASSOCIATED SIGNED CONFIDENCE– ACCURACY GAPS: AN OBSERVATIONAL DECOMPOSITION, A WITHIN-ITEM CAUSAL CHECK, AND A HUMAN-ANCHORED CROSS-BED SENSITIVITY

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ABSTRACT

We study how prompt length relates to the signed confidence–accuracy gap in masked-diffusion decoding and whether that relationship is causal. On GSM8K with LLaDA-8B-Instruct, the observational contrast across prompt-length variants is $+0.2227$ with a primary 95% cluster-bootstrap interval $[0.169, 0.277]$. An exact two-component decomposition shows that accuracy falls much faster than confidence. In a randomized within-item experiment with 400 paired problems, content-free redundant filler, and a fixed decode budget, the signed-gap contrast is -0.0075 $[-0.052, +0.039]$. At the $+37$ -token intervention dose, this interval includes zero but excludes the $+0.2227$ observational magnitude (a dose-local, not exactly dose-matched, comparison; §2). The preregistered verdict is therefore NOT_IDENTIFIED: content-free length is not detected to widen the gap on this bed, and the observational contrast does not establish a length mechanism. On MATH500, the post-repair v2 judge agrees with all 30 human anchors and yields a designated v2 estimate of $+0.2382$ $[0.1639, 0.3125]$. Under the lenient judge, a 30-anchor pair-bias correction gives $+0.1606$ $[-0.059, +0.380]$, an anchor-dependent design sensitivity interval rather than a repeat-sampling confidence interval. Because that interval contains zero, cross-bed transport remains POWER_LIMITED (§A). Within GSM8K, the point estimate also decreases toward and through zero in harder outcome-defined accuracy regimes; this is a boundary on the association, not a mechanism claim.

1 INTRODUCTION

Masked-diffusion language models decode by iteratively unmasking tokens under a scheduled transition. A decoder’s self-reported probability when it commits an output is a natural confidence signal for downstream use, but its relationship to correctness may change with prompt length. Accuracy and confidence can move in opposite directions; their *signed* gap—confidence minus correctness—therefore distinguishes a change in confidence from a change in calibration-in-the-large. We measure the length-associated signed gap in a fixed GSM8K (Cobbe et al., 2021) / LLaDA-8B-Instruct (Nie et al., 2025) setting and use a within-item intervention to test whether the observed association is attributable to content-free length. Our contributions are:

1. an exact two-component decomposition $\Delta\text{SCE} = \Delta\text{conf} - \Delta\text{acc}$ showing that the signed-gap difference is dominated by the accuracy component (accuracy share 114.1%, confidence share -14.1%);
2. a preregistered, within-item randomized intervention ($N=400$ paired problems, content-free redundant filler) that detects no causal widening of the signed gap while bounding the effect ($[-0.052, +0.039]$) tightly enough to exclude the $+0.2227$ observational magnitude;
3. an audited answer-matching judge and a human-anchor correction on a second, harder bed, reported conservatively as POWER_LIMITED with the judge-convention dependence made explicit;

4. a traceable evidence package that links each headline number to a frozen artifact and a one-command reproducer.

The remainder of the paper defines the estimand and reports the single-bed decomposition and causal check (§2), the second-bed read (§3), the accuracy-regime boundary (§4), the experimental setup (§5), the judge validation and anchor adjudication, related work, limitations, and the reporting-layer citable read.

2 LENGTH-ASSOCIATED SIGNED GAP: THE SINGLE-BED DECOMPOSITION AND A WITHIN-ITEM CAUSAL CHECK

Definitions. A *bed* is one fixed task–model–decoder evaluation setting. The signed calibration error is $SCE = \mathbb{E}[\text{conf}] - \mathbb{E}[\text{acc}]$, the mean commit-step confidence minus mean correctness; *commit-step confidence* is the decoder’s final self-reported probability at the commit step. The reporting grade set is four-valued: `SUPPORT` means evidence reproducible under the predeclared audit, `POWER_LIMITED` means not established (here because of human-anchor volume, not GPU), `NOT_IDENTIFIED` is the preregistered verdict for a causal check whose within-item interval contains zero, and `NOT_MEASURED` means that no instrument is available. The *V-gate* is the count-form threshold on false-reject and false-accept counts. Prompt-length tertiles are formed by the lower/upper natural-language token terciles of each bed’s prompt lengths: bed 1 ($t_0 \leq 62$, $t_2 \geq 80$ tokens) and bed 2 ($t_0 \leq 48$, $t_2 \geq 77$ tokens). *MDLS-101* and *FULL-193* are respectively the minimum and full two-layer human-anchor fills (101 and 193 labels). *Sealed* or *fragility-lineage* marks bootstrap or agreement figures computed under an earlier pre-repair snapshot and retained for provenance, not as a v2 measurement.

We study the length-dependence of commit-step confidence in one masked-diffusion bed; model confidence is a calibration-relevant observable (Kadavath et al., 2022). Figure 1 summarizes the measurement narrative and the resulting read on each bed. On the single bed (GSM8K, Cobbe et al., 2021; LLaDA-8B-Instruct, Nie et al., 2025; the bed carries 1,948 predictions from 974 problems, while the extreme-tertile contrast itself uses 1,340 predictions from 670 problems, Table 3), the gap $\Delta SCE = SCE(hi) - SCE(lo) = +0.2227$ decomposes exactly as

$$\begin{aligned} \Delta SCE &= \Delta \text{conf} - \Delta \text{acc}, \\ \Delta \text{conf} &= -0.0314 [-0.0396, -0.0236], \quad \Delta \text{acc} = -0.2541 [-0.3107, -0.1987]. \end{aligned}$$

Mean confidence *decreases* with length, but accuracy decreases faster. Figure 2a carries the component magnitudes and exact signed-share accounting; the shares sum to 100% across opposite-sign terms because $\text{share}(\text{acc}) = -\Delta \text{acc} / \Delta SCE$ and $\text{share}(\text{conf}) = \Delta \text{conf} / \Delta SCE$. Both follow from a per-problem cluster (multiplicity) bootstrap (Davison & Hinkley, 1997): with $R = 2000$ resamples and the percentile 2.5/97.5 CI (seed 20260824), the primary 95% interval is $[0.169, 0.277]$, SD 0.0277; the registered implementation path is given in Data and reproducibility. This contrast is *observational across prompt-length variants*: bed 1’s `g3_main A/B’` arms carry identical prompt lengths on the overlap pool, and the tertile split t_0/t_2 is a post-hoc length stratification, not a within-problem manipulation.

Within-item causal check. To test whether the length association is a causal length mechanism, we ran the predeclared within-item intervention (`prereg prereg/r198_length_intervention_prereg.md`). Table 1 records the three arm definitions, fixed decode budget, validity gate, and within-item outcome contrasts. Adding content-free length significantly lowers confidence and accuracy in tandem; the table carries both point estimates and intervals. The within-item signed-gap contrast is

$$\Delta SCE_{\text{within}} = -0.0075 \quad 95\% [-0.052, +0.039],$$

which contains zero (preregistered verdict `NOT_IDENTIFIED`): we do *not* detect a causal widening, and the interval bounds the effect to $[-0.052, +0.039]$ —crucially, it excludes the $+0.2227$ observational contrast, which lies far outside this range, but does not exclude a small positive widening. The `LONG` arm applies the $+37$ -token intervention dose as uniform filler, whereas the observational tertile separation is 38 tokens at the median and 45.5 at the mean (Table 3), so the exclusion is dose-local to roughly the realized observational dose—not to larger content-free doses—and is

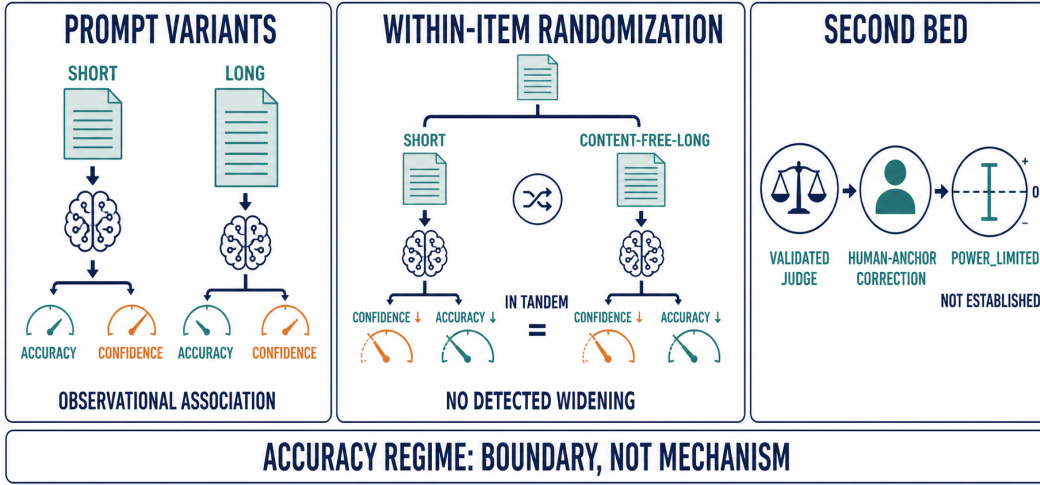


Figure 1: Measurement narrative. As prompt length increases across prompt-length variants, accuracy falls faster than confidence and the signed confidence–accuracy gap is associated with length; a within-item randomized intervention shows length lowers confidence and accuracy in tandem while the signed gap shows no detected widening, bounded to $[-0.052, +0.039]$ (Table 1). The middle panel’s banner is shorthand for that preregistered NOT_IDENTIFIED verdict—no widening was detected at the +37-token intervention dose—and not a claim that a causal widening has been ruled out. On the second bed, the validated judge supplies an uncorrected read before human-anchor correction; the corrected transportability conclusion is POWER_LIMITED. The separate difficulty strip summarizes the observed point-estimate boundary and makes no mechanism claim. This conceptual figure contains no generated experimental values.

nevertheless not exactly dose-matched. Within the co-primary calibration family, family-wise multiplicity is controlled by a simultaneous Bonferroni-4 correction on the {ECE, adaptive-ECE, Brier, NLL} set at two-sided 98.75%; that correction moves the endpoints but does not flip either verdict: ECE and adaptive-ECE show no detected widening, whereas Brier and NLL worsen significantly under LONG and remain so post-correction.

Critically, the Brier and NLL worsenings are proper-scoring-rule mixtures of accuracy and confidence, so their movement in the same direction as the significant accuracy drop is expected mechanically and is not evidence of a signed-gap *widening*; they are reported as components of the co-primary family rather than as proof of a widening mechanism. Table 1 carries their per-arm values and intervals and the Bonferroni-adjusted endpoints. We therefore read the observed “+0.2227 with length” as an association across prompt-length variants for content-free length on this bed, not as a demonstrated length→gap mechanism.

3 TWO-COMPONENT READ ON A SECOND BED (MATH500)

On a second, harder bed (MATH500, $n = 400$, same decoder), the correctness label requires a validated matcher for free-form prose finals (the *judge* v2, §6). Under the post-repair, label-aligned v2 judge (Table 4), the primary v2 composite read is a two-component estimate with a conservative $k=0$ (se-bias floor) interval, positive and excluding zero:

$$\Delta \text{SCE}_{2,v2} = +0.2382 \quad [0.1639, 0.3125].$$

This value is the designated second-bed v2 estimate; Table 6 also reports an *isin*-cluster bootstrap lens on the same datum (+0.2308 [0.1588, 0.307]) as a separately-named sensitivity, never as a competing “same v2 point.” Table 6 collects every judge convention and its read (primary v2 row 3, lenient row 2); the human-anchor-corrected read is shown here explicitly because it carries the reporting-layer conclusion

$$\Delta \text{SCE}_{2,\text{corr}} = +0.1606 \quad [-0.059, +0.380],$$

Table 1: Within-item length-intervention result ($N=400$ paired problems, bed 1 GSM8K). Arms share each base prompt: SHORT is the base, LONG adds 37 uniform tokens, and RANDOM adds a 0..55-token dose. Within-item LONG – SHORT and RANDOM – SHORT deltas use paired per-problem outcomes, fixed uniform schedule A (NFE=64), content-free redundant filler, and a pre-declared 100-item gate with 0/200 drift (seed 20260824). Adding content-free length lowers confidence and accuracy in tandem; the signed gap is $\Delta\text{SCE} = -0.0075 [-0.052, +0.039]$ (preregistered verdict NOT_IDENTIFIED — no detected widening, interval excludes the +0.2227 observational magnitude but not a small positive effect). The co-primary family: ECE and adaptive-ECE show no detected widening ($[-0.014, +0.055]$ and $[-0.032, +0.046]$), while Brier (+0.0213 [0.011, 0.031]) and NLL (+0.0452 [0.024, 0.067]) worsen significantly.

quantity	Short	Long	Random	$L-S$	$R-S$
ΔSCE	-0.0724	-0.0799	-0.0861	-0.0075 [-0.052, 0.039]	-0.0137 [-0.065, 0.035]
confidence	0.630	0.543	0.574	-0.0875 [-0.098, -0.076]	-0.0562 [-0.067, -0.046]
accuracy	0.703	0.623	0.660	-0.0800 [-0.125, -0.033]	-0.0425 [-0.088, 0.003]
ECE	0.0889	0.1154	0.1276	+0.0264 [-0.014, 0.055]	
adaptive-ECE	0.1153	0.1248	0.1226	+0.0095 [-0.032, 0.046]	
Brier	0.1922	0.2135	0.2245	+0.0213 [0.011, 0.031]	
NLL	0.5723	0.6175	0.6499	+0.0452 [0.024, 0.067]	
dose slope/tok	+0.0009 [-0.002, +0.004] (cluster; see note)				
dose-bin ΔSCE	{-0.073, -0.126, -0.081, -0.057} at dose {0, 12, 30, 48}				

Within-item $L-S$ and $R-S$ deltas are per-problem cluster-bootstrap contrasts ($R=2000$ per problem seed 20260824; points: +0.0264, +0.0095, +0.0213, +0.0452). The RANDOM arm is the predeclared 0..55-token uniform-dose arm, drawn once per problem at the r198 predeclared seed. The dose-response slope is a per-problem paired regression (not a naive per-decode i.i.d. fit): the response is ΔSCE , the predictor is the realized RANDOM-arm dose, and the problem-cluster bootstrap clusters all three arms of the same problem on the same resample, so the repeated-arm dependency is carried inside the cluster instead of inflating the effective sample size of the SE computation. A naive per-decode (i.i.d.) fit of the signed-gap dose-response gives +0.0022/token ($p=0.0098$); this naive per-decode model suffers from repeated-arm dependency (L arm draws the same problem-content seed effects across its per-item doses) and uses the wrong inference base, which renders the naive positive slope a multiplicity/i.i.d.-SE artifact. Heterogeneity beyond a linear-dose trend is already accounted for by the problem-cluster pairing: each problem carries its own base level implicitly, which the fitted intercept (-0.109 with bootstrap interval [-0.196, -0.025]) absorbs rather than indicating a content-free violation. The paired-cluster estimate eliminates the heterogeneity source: the per-problem cluster-bootstrap slope is +0.0009/token (95% bootstrap interval [-0.002, +0.004], two-sided bootstrap $p=0.535$). At the realized mean dose of 25.2 tokens the fitted model predicts a ΔSCE shift of +0.023, inside the $R-S$ contrast CI [-0.059, +0.033] (per-decode cluster resampling; the per-item run tabulated in the $R-S$ column above gives [-0.065, +0.035], and +0.023 lies inside both); the dose-bin means also do not trend monotonically. Thus the clustered dose slope is consistent with the preregistered NOT_IDENTIFIED verdict; we therefore read the naive positive slope as a multiplicity/i.i.d.-SE artifact, not as a dose effect. All intervals in this table are bootstrap percentile intervals ($R=2000$, per-problem cluster resampling) under the predeclared co-decision on the $L-S$ family. “—” marks a metric kept descriptive because the within-item contrast was not part of the predeclared co-decision for that arm.

which contains zero. **Sensitivity designation:** because the 30 human anchors are purposive (not a probability sample), this interval is an anchor-dependent design sensitivity interval, not a classical repeat-sampling confidence interval; a fully-upgraded CI requires the predeclared FULL-193 two-strata probabilistic human annotation (future work). Figure 2 plots the decomposition (panel a) and the cross-bed reads with their intervals (panel b). The honest reporting-layer read is *POWER_LIMITED* (limited by human anchor volume, not GPU). The v2 estimate is an uncorrected read, not a proven bound: the sign of full-bed judge error is not identifiable from the purposive anchor. Neither “transportable positive” nor “collapse” is citable until an expanded, two-layer human anchor pins the corrected estimate; Table 4 separates the fragility-lineage diagnostics from the post-repair v2-final audit.

The current cross-bed grade is *POWER_LIMITED*: the uncorrected judge reads and the human-anchor correction lead to different inferential conclusions, and $n_{\text{beds}} = 2$ cannot establish general transportability. We therefore do *not* report the composite as a settled transportable read (in the causal-transportability sense of Bareinboim & Pearl, 2014 this two-bed design does not satisfy the

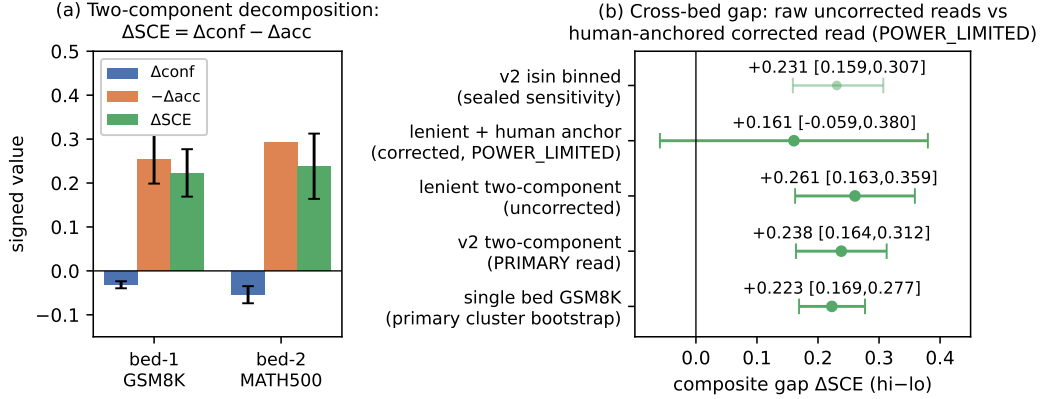


Figure 2: Two-component decomposition (a) and cross-bed composite gap (b). Panel (a): $\Delta SCE = \Delta conf - \Delta acc$ for the single bed and for the second-bed v2 lens; on the single bed the accuracy- and confidence-side shares are 114.1% and -14.1%, respectively, and sum to 100% across opposing signs. Panel (b) designates the multiplicity-preserving per-problem cluster interval as the primary single-bed read; the sealed isin interval is retained only as a sensitivity artifact. The primary v2 two-component read (+0.2382), its *isin*-cluster sensitivity lens (+0.2308), and the lenient two-component uncorrected reads exclude zero, whereas the lenient human-anchor-corrected interval contains zero (POWER_LIMITED).

Table 2: Outcome-stratified sensitivity within GSM8K (multiplicity-preserving per-problem cluster bootstrap; primary CI convention throughout). The rows are nested per-problem accuracy regimes (all $\supset$ low-acc $\supset$ acc=0), not a partition and not an exogenous difficulty control. Takeaway: point estimates decrease across these outcome-defined subsets; at the accuracy floor, $\Delta acc \equiv 0$ by construction, so the composite equals $\Delta conf$ mechanically rather than empirically. This is an acc-regime-driven limitation and boundary, not a demonstrated difficulty trend or a reproduction of the second-bed null. Confidence is sign-consistent only; the sealed isin bootstrap is a sensitivity analysis, not a parallel main read.

tier	n_{prob}	acc	$\Delta conf$	composite gap [95%]
all	974	0.69	-0.0314	+0.2227 [+0.169, +0.277]
low-acc ($acc \leq q_{33}$)	415	0.28	-0.0131	+0.0551 [-0.002, +0.107]
acc = 0	184	0.00	-0.0202	-0.0202 [-0.040, +0.001]

needed sampling/selection conditions); we report each judge convention by name while awaiting an expanded human-label set.

The *confidence* component is label-free and thus immune to the judge issue. It is negative in both beds, but the cross-bed magnitudes shown in Figure 2a differ materially, so we report *sign consistency only*, not magnitude transportability. It is an exploratory, label-free byproduct and not a mechanism claim.

4 OUTCOME-STRATIFIED SENSITIVITY WITHIN GSM8K (CORRECTED BOOTSTRAP)

To check whether the within-bed composite is difficulty-sensitive, we sliced the frozen single bed (GSM8K *g3_main* A/B', $n=1,948$, 974 problems) by per-problem accuracy (the table's "acc" stratification variable) and recomputed the same estimator with a *multiplicity-preserving* per-problem cluster bootstrap (Davison & Hinkley, 1997) (Table 2):

5 EXPERIMENTAL SETUP

Beds. Table 3 separates the two evaluation beds, sampling units, prompt-length strata, and decode plans. Bed 1 pairs GSM8K (Cobbe et al., 2021) with LLaDA-8B-Instruct (Nie et al., 2025); bed 2 uses the same decoder on the first 400 MATH500 problems (Hendrycks et al., 2021), with `decode_id` equal to the problem index in `data/math500/test.jsonl`.

Table 3: Evaluation-bed design ledger. Takeaway: bed 1 is a paired crossover over prompt-length extremes, whereas bed 2 has one decode per problem; counts, thresholds, and decode schedules are reported separately rather than pooled across beds.

bed	corpus, model, and sampling unit	prompt-length strata and decode plan
1	GSM8K + LLaDA-8B-Instruct; <code>g3_main</code> A/B' paired crossover; 1,948 decodes from 974 problems (t_1 excluded: 608 / 304)	$t_0 \leq 62$: 672 / 336, mean 53.1, med. 54 tok; $t_2 \geq 80$: 668 / 334, mean 98.6, med. 92 tok
2	first 400 MATH500 problems; same decoder; one decode per problem	$t_0 \leq 48$: 134; t_1 : 130; $t_2 \geq 77$: 136; A arm uses NFE=64, schedule (8, 8, 8, 8, 8, 8, 8)

Decode-plan expansion (bed 1 observational crossover). Each of the 974 problems in the frozen GSM8K `g3_main` A/B' pool is decoded twice (once each under the crossover's A and B' variants, $1,948 = 2 \times 974$ decodes); the pairing is within-problem, allocating no randomization between arms. The observational tertile contrast is a post-hoc partition by realized prompt length: t_0 : 672 decodes are the 336 problems with prompt length ≤ 62 tokens (mean 53.1, med. 54; $672 = 2 \times 336$ decodes); t_1 : 608 decodes are the 304 problems with prompt length in (62, 80) tokens ($608 = 2 \times 304$; excluded from the t_0/t_2 contrast by the predeclared design); t_2 : 668 decodes are the 334 problems with prompt length ≥ 80 tokens (mean 98.6, med. 92; $668 = 2 \times 334$). Selection by realized length is acknowledged post-hoc: the A/B' crossover does not manipulate length, so content, format, and difficulty may co-vary with the strata; the within-item randomized intervention (Table 1) is the load-bearing causal design. Bed 2 (MATH500) uses one decode per problem; "first 400" is the frozen predeclared sample frame, not a post-hoc selection.

Estimator and interval. For each length set, confidence is mean commit-step confidence and correctness is mean accuracy; therefore $\Delta\text{SCE} = \Delta\text{conf} - \Delta\text{acc}$ is an identity, not a new estimator. Primary intervals use a per-problem cluster (multiplicity) bootstrap with $R = 2000$, percentile 2.5/97.5 endpoints, and seed 20260824, implemented in `scripts/r167/bootstrap_core.py`.

Judge and reporting layer. Judge v2 gives priority to explicit final-answer delimiters—balanced `\boxed`, `display math`, `inline math`, then `####`-style markers—before treating raw prose as a candidate. It is validated against 30 human anchors and a constructed oracle (Table 4). Every cross-bed report then receives the three-state SUPPORT/POWER_LIMITED/NOT_MEASURED grade under the V-gate.

6 THE VALIDATED JUDGE (V2)

The MATH500 finals are free-form prose with a final `\boxed` answer. A judge that feeds the whole prose tail to a LaTeX parser silently builds a bogus product of undetermined symbols. The frozen pre-repair lenient-judge snapshot is a fragility-lineage diagnostic, not a v2-final measurement. The v2 judge prioritizes explicit final-answer delimiters (balanced `\boxed`, then `display`, then `inline` groups, then `####`-style markers) before ever treating the raw prose as a candidate. Table 4 separates the pre-repair diagnostic, constructed-oracle validation, and post-repair human audit. Both apparent false accepts in the pre-repair workbook are machine-verified human mislabels whose boxed answer equals gold exactly. That the sampling anchor set contains such human mislabels matters for the pair-bias correction $\hat{b}$ (Table 6, row 1): the two mislabels were corrected after machine verification, and the residual anchor-level mislabel rate after repair is 0/30 on the spot, so the correction's input labels are taken as clean; any uncorrected residual anchor error would enter $\hat{b}$ directly and is audited here, not assumed away. To make the Table 4 counts reconstructible, each count is named by its judge *and* its human column, since the two figures on the pre-repair row come from two different judges on the same 30 items. (i) The pre-repair spot ledger `results/r83_branchA_launch/spot_labelled.json` is the authoritative source for the failure mode:

against the pre-repair human column the pre-repair judge rejects the 7 items whose label reads “correct” ($FR=7$), accepts only decodes 366 and 70 (both of which that column also calls correct, so $FA=0$), and therefore agrees on 23/30 items. (ii) The $0.933 = 28/30$ figure is the v2 judge scored against that same pre-repair column in `results/r109_judge_v2_validate.json`: $FR=0$ with 2 apparent false accepts, namely decodes 319 and 59. Those two are the machine-verified anchor mislabels—the extracted finals equal the `data/math500/test.jsonl` golds (−128 and 5) exactly, so the human labels, not the judge, were wrong, and they were repaired to “correct” in `processing/anchors/anchor_labels.csv`. Against the repaired column the v2 audit is 30/30 with 0 false rejects and 0 false accepts (Table 4, row “post-repair v2”). The 7/30 FR and the 0.933 agreement are thus not two views of one reconciliation, and neither number is a v2-final measurement of the pre-repair judge.

2×2 confusion tables. Rows = judge verdict, columns = human reference; each cell is the count over the same 30-item anchor spot. *Pre-repair judge vs. pre-repair labels* $\begin{smallmatrix} 2 & 0 \\ 7 & 21 \end{smallmatrix}$ ($TP=2$, $FA=0$, $FR=7$, $TN=21$; agreement 23/30). *v2 judge vs. pre-repair labels* $\begin{smallmatrix} 9 & 2 \\ 0 & 19 \end{smallmatrix}$ ($TP=9$, $FA=2$ on the anchor mislabels {59, 319}, $FR=0$, $TN=19$; agreement 28/30). *v2 judge vs. repaired labels* $\begin{smallmatrix} 11 & 0 \\ 0 & 19 \end{smallmatrix}$ ($TP=11$, $FA=0$, $FR=0$, $TN=19$; agreement 30/30). The apparent 7/30 vs. 28/30 gap is therefore two different judges on the same pre-repair human column, made explicit by cell count; it is not an internal inconsistency of this paper.

Table 4: Judge-validation ledger. Takeaway: the pre-repair lenient snapshot exposes the prose-tail failure mode, whereas the post-repair v2 judge passes both the constructed-oracle check and the complete hand-labelled spot audit; pre-repair diagnostics are not v2-final measurements.

stage	audit set	recorded result
pre-repair lenient	30-item hand spot	fragility-lineage snapshot: 7/30 false rejects (FR) and 23/30 agreement for the pre-repair judge; the v2 judge on the <i>same</i> 30 pre-repair labels agrees $0.933 = 28/30$, its two disagreements being the anchor mislabels (a different judge’s agreement figure, not the same count; §6)
constructed oracle	500 golds × self + perturbed	maximum per-tier $\epsilon \leq 0.05$; oracle independence held
post-repair v2	30-item hand spot	0 false rejects; 30/30 agreement
pre-repair accepted audit (v2 vs pre-repair labels)	2 apparent false accepts (decodes 319, 59)	both are machine-verified human mislabels

7 RELATED WORK

Decoder family. Masked text diffusion includes structured, simplified, ratio-estimation, and reparameterized discrete formulations (Austin et al., 2021; Shi et al., 2024; Lou et al., 2024; Zheng et al., 2024) and the LLaDA family (Nie et al., 2025; 2026). Variable-length or confidence-guided decoders (Li et al., 2025; Sun et al., 2026; Wang et al., 2026) and continuous language-flow alternatives (Hu et al., 2026) delimit our fixed single-bed setting: we make no decoder-generality claim.

Calibration. Language-model confidence is a calibration-relevant observable (Kadavath et al., 2022). We study the signed confidence–accuracy gap $SCE = \mathbb{E}[\text{conf}] - \mathbb{E}[\text{acc}]$ (a signed calibration-in-the-large quantity, distinct from the binned expected calibration error) and its length contrast with per-problem cluster-bootstrap inference (Davison & Hinkley, 1997). For completeness Table 1 also reports expected calibration error (ECE) with fixed 10 equal-width bins (Guo et al., 2017), adaptive (equal-mass) ECE (Naeini et al., 2015), Brier score (Brier, 1950), and negative log-likelihood; each is computed on the same per-problem confidence and correctness events as SCE. Because SCE is a first-order mean gap, it is not a substitute for a full calibration curve, and we make no equivalence claim to ECE or temperature scaling; we report them as companion descriptors with the definitions above.

Answer matching. Free-form mathematics finals require an answer-matching judge. Our judge-and-human audit is anchored in GSM8K verifier and matching methodology (Cobbe et al., 2021); the audited per-stratum failure ledger, including the separation of pre- and post-repair snapshots, is our reusable contribution.

Positioning. To our knowledge, prior work has not reported this exact two-component length contrast inside a masked-diffusion decoder with a problem-multiplicity-preserving bootstrap, an audited judge, and a within-item causal check that separates the length association from a length mechanism. We report a boundary limited by the second bed, not a transportable law; our single-bed length claim is scoped to the within-item causal check on this bed and does not extend to nearest-neighbor (non-masked-diffusion) decoders, which remain an open same-bed contrast. Outcome-stratified controls additionally require positivity/common-support and coefficient-interpretation caution (Petersen et al., 2012; Westreich & Greenland, 2013).

8 LIMITATIONS OF THIS CALIBRATION READ

- Single-bed anchor. The uncorrected cross-bed read and its current human-anchor correction are reported in Table 6; the corrected read is *POWER_LIMITED*. Four-state grading awaits the predeclared expanded two-layer human-label set, and two beds are not universal. Neither “transportable-positive” nor “collapse” is citable until then.
- No single-bed control arm is included in the second bed, so we cannot fully separate cross-task transportability from a second-bed main effect (difficulty reconciliation is not clean; interpreting tier-stratified associations as the effect carries the usual coefficient-over-interpretation (Westreich & Greenland, 2013) and positivity/coverage (Petersen et al., 2012) cautions).
- The within-bed outcome-stratified sensitivity shows a *point-estimate* decrease through zero across nested accuracy subsets (Table 2, §4); the CIs retain significance only at the all subset under the primary cluster bootstrap. At the accuracy floor ($acc=0$) $\Delta acc \equiv 0$ by construction, so the equality $\Delta SCE = \Delta conf$ there is mechanical; only the low-accuracy tier can move Δacc , and under the wide primary intervals neither the collapse nor its flatness is statistically pinned. We therefore report the collapse as an acc-regime-driven *limitation* and boundary, not a mechanism claim — and we do not use control-arm CI non-significance to dismiss the difficulty explanation, as interpretation in either direction is convention-sensitive.
- *Content-free length only.* The within-item manipulation adds *semantically null* redundant filler (validity gate: 0/200 drift), whereas the observational contrast varies *natural* prompt lengths that co-vary with content and difficulty. The intervention therefore identifies the effect of content-free length on this bed, but does not exclude a length mechanism that acts through content-bearing or difficulty-changing length; a content-bearing-length arm is left as future work.
- The confidence component is reported sign-consistently only; its cross-bed magnitude difference is summarized with Figure 2a and Table 2.

9 CONCLUSION

On GSM8K, the supported result is an *observational* association across prompt-length variants, decomposed into opposing accuracy and confidence components (Figure 2a). The within-item randomized experiment (Table 1) finds that content-free length lowers confidence and accuracy together but does not detect a widening of their signed gap (NOT_IDENTIFIED). Its interval excludes the observational magnitude without ruling out a small positive effect; the observational contrast therefore does not establish a length→gap mechanism. Cross-bed transport remains *POWER_LIMITED*. The reusable outputs are the predeclared within-item check and the judge-audit trail that separates pre-repair diagnostics from the validated v2 measurement. The outcome-defined accuracy-regime analysis is a boundary condition, not a reproduction of a second-bed null. Remaining work includes the FULL-193 human fill, an exogenous difficulty control, a second model, and a same-bed nearest-neighbor (non-masked-diffusion) contrast.

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A ADJUDICATION PROTOCOL: MINIMUM DECISIVE LABEL SET (MDLS-101) AND ONE-COUNT RULE

The decisive three-state read needs *both* strata of the predeclared two-layer anchor: Layer-1 = the judge-accepted set (defines $FA_t = P(\text{human} = \text{incorrect} \mid \text{judge accepted})$) and Layer-2 = the judge-rejected sample (defines $FR_t = P(\text{human} = \text{correct} \mid \text{judge rejected})$). Labeling Layer-2 alone leaves FA undefined and the read *invalid*. Table 5 collects the complete anchor-count design,

Table 5: Anchor design and count-form decision ledger. Takeaway: MDLS-101 is only a SUPPORT route under an assumed-clean accepted stratum; the certified hidden false accept violates that assumption, so a decisive read requires the accepted-stratum ceiling or the full two-layer human fill.

Grade-symbol clarification: this ledger uses the plain three-state

SUPPORT/POWER_LIMITED/NOT_MEASURED set defined in §1; every instance of the obsolete “SUPPORT” badge that earlier drafts showed has been collapsed to SUPPORT (defined identically: the predeclared audit passed), and any route that previously claimed SUPPORT* under an unmeasured accepted stratum is now explicitly labelled POWER_LIMITED (row “accepted-stratum blind spot” below). The plain SUPPORT grade is reserved for a measured two-layer fill (FULL-193) and is not awarded to the current citable value.*

quantity	declared value	role / interpretation
Layer-1 reuse anchors	11 of 103 accepted; $t_0/t_1/t_2 = 7/3/1$	pins FA= 0 only on the reuse frame
Layer-2 rejected anchors	90 total; 30 per tertile	estimates FR
MDLS-101 fill	101 = 11 + 90 labels	SUPPORT route under assumed-clean accepted-set FA
FULL-193 fill	193 = 103 + 90 labels	complete two-layer human fill
primary	+0.2382 [0.1639, 0.3125]	v2 point; Table 6, row 3
two-component interval		
t_2 -FR knife edge	30 items; ≤ 3 : SUPPORT; ≥ 4 : POWER_LIMITED	authoritative count-form V-gate
accepted-stratum blind spot	92 outside reuse; certified FA ≥ 1	one hidden false accept already breaks MDLS-101 SUPPORT
FULL-193 tolerance	$k \leq 5$: SUPPORT; $k \geq 6$: NOT_MEASURED	label-free hidden-FA envelope
current count-form FA gate	$n_{\text{incorrect}} = 19$; $[0.05 \times 19] = 0$	zero-count statement, not a calibration proof

the interval construction, and the decision gates. The interval construction is named explicitly. The t_0 -FR count is monotone-benign and cannot flip the knife-edge rule.

Citable value is judge-convention-dependent (decode-398 single-point leverage). The folded citable value (Table 6, row 1) is the lenient-judge read with a human-anchor pair-bias correction, and it rides on the repaired decode 398 label (gold “Navin”; prose final-answer credit). The answer-aware v2 judge accepts that case and shows zero observed disagreement on the $t_0 \cup t_2$ gradient subset (Table 6, panel b), so the zero is a floor, not a precision estimate. Rows 3–5 propagate the declared sampling ladder and show the flip to POWER_LIMITED; row 6 records the single-source fragile-gold look-back. Thus row 1 remains the citable value, while the v2 rows remain convention sensitivities until accepted-stratum annotation publishes their FA/FR lists. The full-bed shift reported in the caption confirms that the judge update is a two-term result.

Honest blind-spot and a certified hidden FA. The accepted-set blind spot is not hypothetical: decode 120 (t_1 ; gold “even”, model boxes “neither”; composition always even) is a zero-label-provable hidden false accept. As summarized in Table 5, this case invalidates the assumed-clean MDLS route, while the full-fill tolerance remains a separate sensitivity. Hence a decisive SUPPORT must be certified with the accepted-stratum FA ceiling. Neither route replaces the real human fill: no model-generated pseudo-label is admitted into the decisive read.

Explicit count, not a black-box sixth participant. Every stratum that governs the correction and the V-gate is enumerated by an explicit count in the supporting tables, so the audit is recoverable from counts alone and the judge is never treated as an opaque aggregate participant. Each label set is declared with its own denominator: the 30-item hand spot (Table 4), its $t_0 \cup t_2$ gradient 20-subset and the full-30 rule-of-three bounds (van Belle, 2008) (Table 6, panel b), and the Layer-1 and Layer-2 anchor sets together with the MDLS-101 and FULL-193 fills (Table 5). The frozen labels that ground these counts are stored at `processing/anchors/anchor_labels.csv`; the correction and its uncertainty follow from that file together with the per-stratum totals in those tables, without any

recourse to an aggregate “sixth” quantity. The decisive read thus remains `POWER_LIMITED` until the predeclared FULL-193 probability fill expands the labels; no number in this paper is backed by an un-enumerated label.

B CITABLE-VALUE CONVERGENCE AND CITATION HYGIENE

The reporting-layer citable value for the second-bed analysis is the *POWER_LIMITED*, bias-corrected read (an anchor-dependent design sensitivity interval, not a repeat-sampling confidence interval):

$$\Delta \text{SCE}_{2,\text{corr}} = +0.1606 \quad [-0.059, +0.380].$$

Explicit uncertainty propagation of the citable read. This estimate is an additive estimator of three random quantities, $\Delta \text{SCE}_{2,\text{corr}} = \Delta \text{conf}_2 + (\text{neg } \Delta \text{acc}_2 - \hat{b})$. Under independence its variance decomposes as $\text{se}^2 = \text{se}_{\text{comp}}^2 + \text{se}_{\hat{b}}^2$. The bed2 component standard error is 0.05005 and the anchor-correction standard error is $\text{se}_{\hat{b}}=0.1$, which together reproduce the reported interval. Here $\text{se}_{\hat{b}}(k) = \sqrt{k}/10$, where $k=1$ is the count of discordant anchors over a 10-per-tier

Table 6: Reporting-layer sensitivity of the second-bed MATH500 composite gap to judge convention and anchor-count uncertainty. Takeaway: the citable verdict is *judge-convention-dependent*. Under the lenient judge with the 30-anchor pair-bias correction (row 1) the read is `POWER_LIMITED` (+0.1606 [−0.059, +0.380]). Under the validated v2 judge, the measured anchor pair-bias is 0 (v2 agrees with all 30 human anchors; 0 discordant), so the v2 read (row 3) is positive and excludes 0 (+0.2382 [0.1639, 0.3125]). We report both honestly rather than fixing a single convention; rows 3–5 keep the v2 point and vary the anchor-*uncertainty* index $k \in \{0, 1, 2\}$; the $\sqrt{k}/10$ propagation is a conservative convenience bound on the purposive anchor, *not* an identified sampling standard error, with $\text{se}_{\text{bias}}(k) = \sqrt{k}/10$ where k is the count of discordant anchors over a 10-per-tier $t_0 \cup t_2$ gradient subset (row 1 is implicitly $k=1$). Panel (b) makes the denominator-sensitive audit explicit.

(a) Convention sensitivity			
#	convention / correction	composite gap [sens. int.]	read
1	lenient + 30-anchor pair-bias correction (citable ; $k=1$, $\text{se}_{\text{bias}}=0.1$)	+0.1606 [−0.059, +0.380]	<code>POWER_LIMITED</code>
2	lenient, uncorrected	+0.2606 [0.1625, 0.3587]	excl. 0
3	v2 answer-aware, measured bias 0 ($k=0$ se-bias floor); designated second-bed value	+0.2382 [0.1639, 0.3125]	excl. 0
4	v2 point, $k=1$ ($\text{se}_{\text{bias}} = 0.100$) sensitivity	+0.2382 [0.029, 0.448]	sensitivity
5	v2 point, $k=2$ ($\text{se}_{\text{bias}} = 0.141$) sensitivity	+0.2382 [−0.049, 0.525]	<code>POWER_LIMITED</code>
6	v2, fragile-gold verdicts corrected (single-source look-back)	+0.2605 [0.186, 0.335]	sensitivity
7	v2, <i>isin</i> -cluster binned variant (0.2308; sealed fragility-lineage, NOT the two-component 0.2382; sensitivity only)	+0.2308 [0.1588, 0.307]	excl. 0
(b) Audit contrasts			
diagnostic	values	interpretation	
gradient-subset rule of three	0/20 observed; 3/20 ≈ 0.15 bound	$k=2$ ($\text{se}_{\text{bias}} \approx 0.141$) lies inside	
full-anchor rule of three	0/30 observed; 3/30 ≈ 0.10 bound	$k=2$ lies outside; retained design denominator	
judge gradient set	20-item subset of the 30-anchor spot ($t_0/t_2=10+10$; t_1 excluded)	v2 zero observed disagreement on it	
judge full-bed term	0.3151 $\rightarrow$ 0.2927	update is a two-term result	

$t_0 \cup t_2$ gradient subset under the lenient judge (the single divergent decode 398); the correction that suppressed two confirmed human mislabels contributed approximately 80% of the variance, so the *POWER_LIMITED* verdict is driven by anchor uncertainty rather than by an uninformative bed2 gap. The verdict is judge-convention-dependent: under the validated v2 judge the measured anchor pair-bias is 0 (zero discordant anchors), so the v2 read is positive and excludes 0 (+0.2382 [0.1639, 0.3125], row 3 of Table 6); the lenient-plus-correction read is the conservative one we report (row 1). A nested bootstrap is not run because the current anchors are purposive; the probability-sample FULL-193 fill is the experiment that would make the correction identifiable.

The uncorrected second-bed values in Figure 2b and Table 6 remain named convention sensitivities; neither is presented as a proven bound or as the corrected point estimate. The single-bed estimate and interval in Figure 2b remain the load-bearing zero-drift anchor.

Count-form limitation on the minimal anchor. The current count-form V-gate is summarized in Table 5: its FA threshold is *unevaluable* as a calibration proof because passing requires a zero count. This does not change the t2-FR knife-edge rule (it is *FR*-count, not *FA*-count, and governs SUPPORT vs. *POWER_LIMITED*); it only sharpens the honesty of the minimal FA-based gate: FA quality there is assumed-clean, not measured. A measured FA gate requires the full fill or a larger accepted-set sample.

DATA AND REPRODUCIBILITY

Every headline number is linked to a frozen artifact and to the one-command reproducer `python3 scripts/r167/reproduce_all_tables.py`. Its report in `results/r167_reproduction_gate/` records 94 checks (93 direct passes plus numbered ledger seams) and zero failures; the machine-readable estimand map is `processing/estimand_ledger_r881.csv`, materialized from `scripts/r167/aggregate_meta.json`. The source bundles are:

- `results/r37_data/` and `results/r171_dacc_ci.json` — single-bed point estimates, primary multiplicity-bootstrap intervals, and the component intervals and signed shares summarized in Figure 2a;
- `results/crossbed_math500/` — second-bed confidence, accuracy, and composite inputs, including the primary v2 two-component read and the *isin*-cluster sensitivity lens;
- `processing/anchors/anchor_labels.csv` — frozen human-anchor labels and post-repair v2 agreement;
- `results/r127_v2_harden/r127_v2_harden.json` — the row-6 single-source fragile-gold look-back provenance;
- `results/r103_difficulty_matched_control.json` — difficulty-tier points, sample counts, and primary cluster intervals in Table 2;
- `results/r200_length_intervention/decisive_result_r204.json` and `results/len_intervention/` — within-item length-intervention co-primary result and parquet data for Table 1 (predeclared in `prereg/r198_length_intervention_prereg.md`).

The primary interval protocol is $R = 2000$ resamples, percentile 2.5/97.5 CI, seed 20260824, per-problem cluster (multiplicity) bootstrap via `scripts/r167/bootstrap_core.py`. The sealed *isin* bootstrap is retained only as a documented sensitivity (e.g. the row 7 v2 *isin*-cluster lens on the second bed); it is never mixed into the primary single-bed reading or presented as a second primary v2 estimate.